

head $p_{\phi_t}(z_t|z)$, and the corresponding prediction variable $\hat{Y}_t$ is non-parametric mapping of Z_t . Then, define information flow (Shwartz-Ziv and Tishby 2017; Goldfeld et al. 2019) in neural networks as a Markov chain shared by all tasks, i.e., $Y_t \rightarrow X \rightarrow Z \rightarrow Z_t \rightarrow \hat{Y}_t$ for task t .

Methodology

To ensure sufficiency for target tasks and mitigate the negative effects of redundant features, we propose a principled multi-task representation learning framework (InfoMTL) that extracts noise-invariant sufficient representations for all tasks. It contains two learning principles: shared information maximization (SIMax) and task-specific information minimization (TIMin), which constrain the amount of shared and task-specific information in the multi-task learning process.

Shared Information Maximization Principle

In the MTL paradigm, the mutual information between the input X and the shared representations Z is typically reduced implicitly (Figure 1(a)) or explicitly (Figure 1(b)) under the information bottleneck (IB) theory (Shwartz-Ziv and Tishby 2017; Kawaguchi et al. 2023). However, directly reducing redundancy for one task is prone to losing necessary information for others. As a result, the learned shared representations Z usually suffer from task-specific insufficiency.

To alleviate the insufficiency issue, a shared information maximization (SIMax) principle is proposed to learn more sufficient shared representations for all target tasks. The SIMax principle simultaneously maximizes the mutual information between the input X and the shared representations Z for all tasks, as well as the mutual information between the shared representations Z and the target task Y_t for each task t . It can be formulated as the maximization of the following Lagrangian,

$$\max \sum_{t=1}^{|T|} [I(Y_t; Z)] + \alpha I(X; Z), \quad (1)$$

subject to the Markov constraint, i.e., $Y_t \rightarrow X \rightarrow Z \rightarrow Z_t \rightarrow \hat{Y}_t$. α is a parameter that balances the trade-off between the informativeness of Z for X and Y_t .

In Equation (1), the second term promotes the task-agnostic shared representations Z preserves as much information as possible about the input X . It ensures the sufficiency of the shared representations Z for all potential targets Y . The first term encourages the shared representations Z to capture the necessary information relevant to the target Y_t for each task t . As shown in Figure 1(b) and (c), unlike IB-based MTL methods (Qian, Chen, and Gechter 2020; de Freitas et al. 2022) that explicitly minimize input-representation information, our InfoMTL with SIMax handles the information in an opposite manner to avoid the insufficiency issue arising from representation compression in the multi-task paradigm.

Implementation of SIMax The implementation of SIMax contains two terms, i.e., maximizing $\sum_{t=1}^{|T|} [I(Y_t; Z)]$ and $I(X; Z)$. Firstly, we maximize the lower bound of $I(Y_t; Z)$

by estimating the conditional entropy of the target Y_t given the shared representations Z . Following Kolchinsky, Tracey, and Wolpert (2019) and Hu et al. (2024a), we use cross-entropy (CE) as the estimator for each classification task t . Secondly, to maximize $I(X; Z)$, we use the InfoNCE estimator (Gutmann and Hyvärinen 2010; van den Oord, Li, and Vinyals 2018) to optimize the lower bound of $I(X; Z)$ during the multi-task encoding process. According to the information flow of $Z \leftarrow X \rightarrow Z'$, the Markov chain rule states that $I(X; Z) \geq I(Z; Z')$. For the InfoNCE estimation of maximizing $I(Z; Z')$, the selection of positive and negative samples as well as the implementation of noise-contrastive loss are consistent with Hu et al. (2024b). The optimization objective of SIMax for MTL can be:

$$\begin{aligned} \mathcal{L}_{\text{SIMax}} = & \mathbb{E}_{z \sim p_{\theta}(z|x)} \{ \mathbb{E}_{t \sim T} [-\log q_{\phi_t}(y_t|z)] \\ & - \alpha \log \frac{\exp(\text{sim}(z, z^+)/\tau)}{\sum_{z' \in \mathcal{B}^+} \exp(\text{sim}(z, z')/\tau)} \}, \end{aligned} \quad (2)$$

where $p_{\theta}(z|x)$ is a shared encoder with parameters θ . $q_{\phi_t}(y_t|z)$ is a task-specific decoder with parameters ϕ_t for task t , and its output distribution is adapted for task prediction by a non-parametric function (e.g., Softmax operation for classification). z^+ refers to the positive key of z , generated by dropout in $p_{\theta}(z|x)$. $\mathcal{B}^+$ represents the set of positive keys in the current batch $\mathcal{B}$. $\text{sim}(\cdot)$ is a pairwise similarity function, i.e., cosine similarity. $\tau > 0$ is a scalar temperature parameter that controls the sharpness of the probability distribution, which is applied during the Softmax operation.

Task-specific Information Minimization Principle

In multi-task scenarios, redundant information often differs across tasks, such that information beneficial for one task may become redundant for another. Directly applying the IB principle (Tishby, Pereira, and Bialek 1999; Tishby and Zaslavsky 2015) to compress the task-specific redundancy for one task can result in the loss of necessary information for other tasks. Consequently, both the shared representations Z and the task-specific output representations Z_t for a given task t often contain redundant features that are irrelevant to the specific task.

To better alleviate the redundancy issue, a task-specific information minimization (TIMin) principle is designed to mitigate the negative effect of potential redundant features in the inputs for the target task. It can compress task-irrelevant redundant information and preserve necessary information relevant to the target for multi-task prediction. The TIMin principle minimizes the mutual information between the input X and the task-specific output representations Z_t , while maximizing the mutual information between the output representations Z_t and the corresponding target task Y_t for each task t . The principle can be formulated as the maximization of the following Lagrangian,

$$\max \sum_{t=1}^{|T|} [I(Y_t; Z_t) - \beta I(X; Z_t)], \quad (3)$$

subject to $Y_t \rightarrow X \rightarrow Z \rightarrow Z_t \rightarrow \hat{Y}_t$ for each task t . β is a trade-off parameter of the compression of Z_t from the input X and the informativeness of Z_t for Y_t .